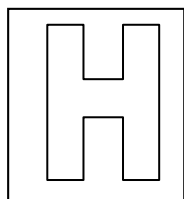


Candidate Name: _____

Class Adm No

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2018 End-of-Year Exams Pre-University 2

H1 Biology

8876/02

Paper 2 Core Paper

10 September 2018

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	/ 11
2	/ 8
3	/ 7
4	/ 10
5	/ 9
Section B	/ 15
Total	/ 60

This question paper consists of 14 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. The process of DNA replication is sometimes described as “*asymmetrical*” replication as the daughter strands are synthesised slightly differently in the form of a leading strand and a lagging strand. Fig. 1.1 shows a replication bubble and the origin of replication.

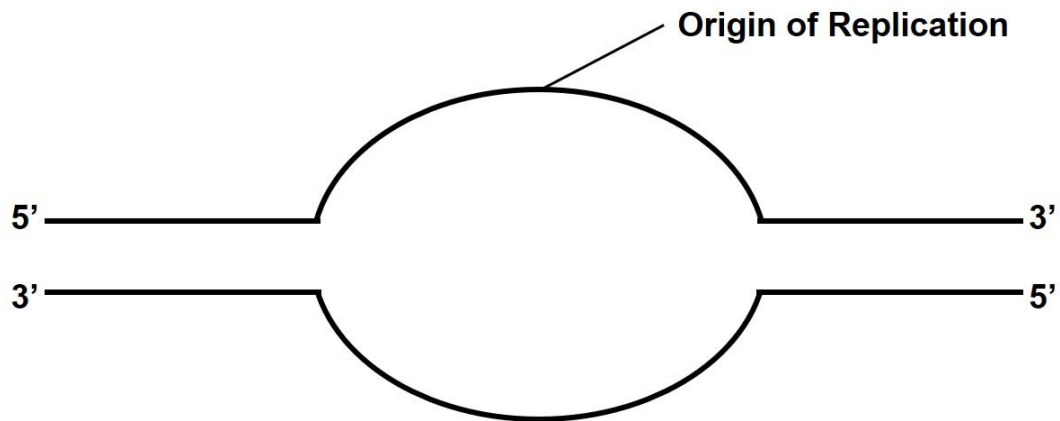


Fig. 1.1

(a) (i) On Fig. 1.1, draw the following with respect to one of the replication fork:

- Leading and lagging strands
- Arrow heads ($\rightarrow$) on leading and lagging strands to indicate direction of synthesis
- 5' and 3' ends of leading and lagging strands

[2]

(ii) Explain why the two daughter strands are synthesised asymmetrically.

.....

.....

.....

.....**[2]**

Telomeres are short repetitive sequences found at the ends of chromosomes. Length of telomeres can be used to estimate cellular age as telomeres shorten over time.

(b) State what causes telomeres to shorten over time.

.....
[1]

A single human diploid cell contains around six billion base pairs, amounting to around two meters in length. The diameter of a nucleus is around 10^{-15} meters.

(c) Describe how DNA are packaged such that it can fit into the nucleus.

.....

[3]

Senescence cells are cells that have telomeres that have shortened to a critical length such that the cells no longer undergo cell division.

(d) (i) State the cell cycle stage that senescence cells are in.

.....[1]

(ii) Explain the purpose for limiting the number of times a cell can divide.

.....

[2]

[Total: 11]

2. Rubisco is an enzyme catalysing the first stage of Calvin cycle, its function is vital in maintaining life on Earth.

(a) (i) State the initial reactants and final product(s) of Calvin cycle.

.....
[1]

(ii) Explain how Rubisco plays a role in maintaining life on Earth.

.....

[2]

Rubisco gene is found in all photosynthetic plants. A study was carried out to investigate the effect of temperature on the catalytic activity of Rubisco found in different plants. Fig. 2.1 shows the results.

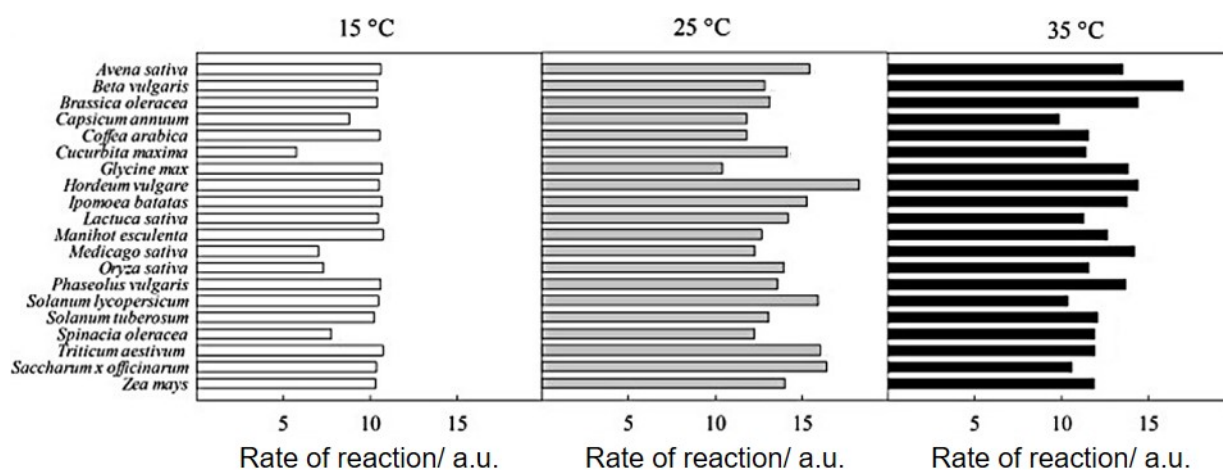


Fig. 2.1

- (b)** With reference to Fig. 2.1, conclude if Rubisco found in different plants have the same optimal temperature range.

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.....[3]

- (c)** Explain why Rubisco from different plants exhibit different catalytic activity under the same temperature condition with all other conditions kept constant.

.....

.....

.....

.....[2]

[Total: 8]

3. In *Drosophila melanogaster*, the common fruit fly, wing shape is controlled by a gene with multiple alleles. These alleles are listed below in no particular order. When pure-breeding *Drosophila* with long wings were crossed with pure-breeding *Drosophila* with vestigial wings, all offspring had long wings.

Long = W^L

Vestigial = W^V

Antlered = W^a

Drosophila with vestigial wings crossed with *Drosophila* with antlered wings produce all offspring with wings of a new phenotype that appears to be an intermediate between vestigial and antlered phenotype, called vestigial-antlered.

(a) Suggest an explanation for the new phenotype.

.....

[2]

Eye colour of *Drosophila* is determined by a gene located on the X chromosome. Sex determination in *Drosophila* is similar to that of humans. *Drosophila* can either have red eyes or white eyes, the allele coding for red eyes is dominant over the allele for white eyes.

Pure breeding female *Drosophila* with long wings and red eyes were crossed with pure breeding male *Drosophila* with vestigial wings and white eyes.

In the F1 generation, all female and male *Drosophila* had long wings and red eyes.

Female and male flies from F1 generation were then crossed to produce the F2 generation.

The observed number of F2 flies with each phenotype were as follows:

Female, long wings, red eyes	595
Female, vestigial wings, red eyes	201
Male, long wings, red eyes	304
Male, long wings, white eyes	299
Male, vestigial wings, red eyes	101
Male, vestigial wings, white eyes	100

- (b)** Draw a genetic diagram to explain the observed results of the cross between female flies and male flies from the F1 generation.

[5]

[Total: 7]

[Turn over]

4. Global temperature records show temperature anomaly ($^{\circ}\text{C}$), which is the difference in temperature of a particular year to the average temperature of a reference year-span. Fig 4.1 shows the global temperature records from 1900 to 2014 relative to the average temperature from 1951 to 1980.

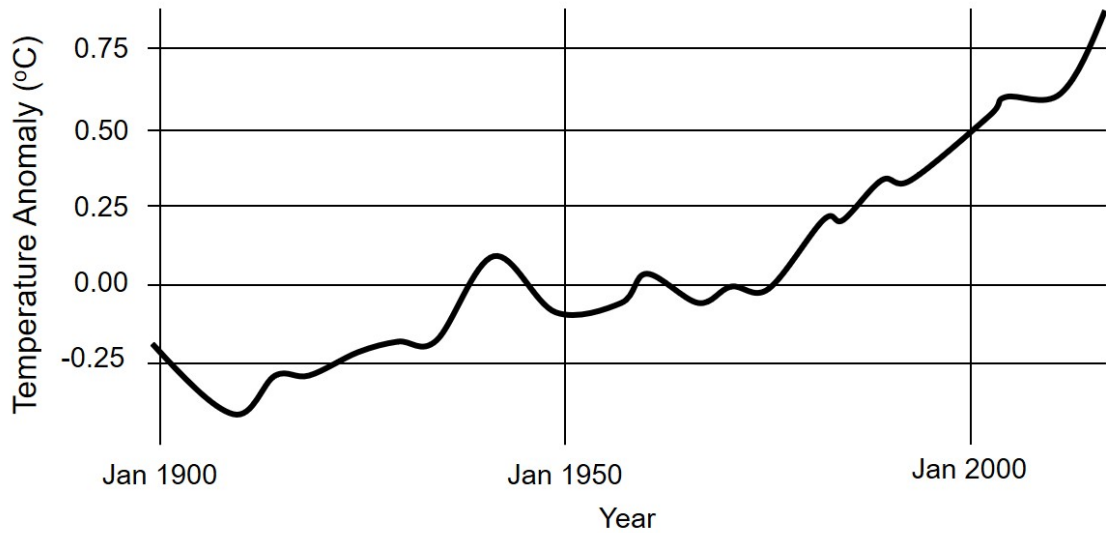


Fig. 4.1

Fig. 4.2 shows the global carbon emission from burning of fossil fuel from 1900 to 2014.

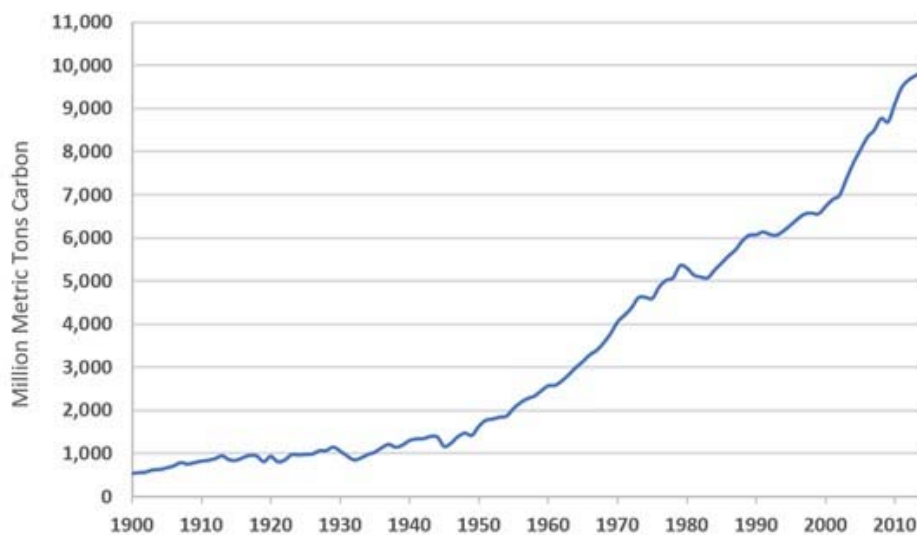


Fig. 4.2

- (a)** Comment on the trend observed for both global temperature records and global carbon emission from 1950s onwards.

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.....[2]

Changes in global temperature will have an impact on many other abiotic and biotic factors, one of them being sea levels.

- (b)** With reference to Fig. 4.1, account for how sea levels have been affected.

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.....[3]

Another factor that is impacted by changes in global temperature is the spread of mosquito borne diseases like Dengue fever (DF), caused by the dengue virus. Fig. 4.3 shows the number of reported DF cases globally from 1980 to 2002.

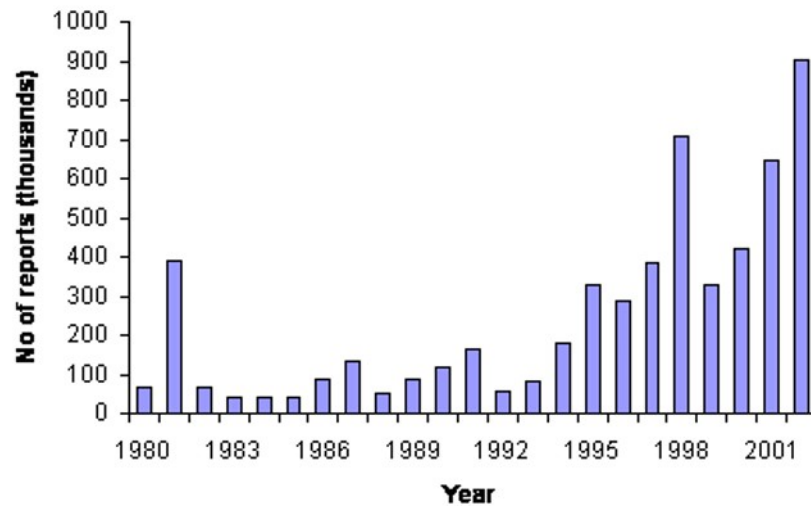


Fig. 4.3

- (c) Explain the relationship between the data in Fig. 4.1 and the general trend observed in Fig. 4.3.

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.....[3]

The Intergovernmental Panel on Climate Change (IPCC) has made predictions on future global surface temperature based on past trends and existing data. If no actions are done, the trend observed in Fig. 4.1 can be expected to continue.

(d) Assuming if the trend in Fig. 4.1 continues, predict how dengue would spread in the future.

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.....[2]

[Total: 10]

5. The length of the upper leg bone is an important variable in research on human evolution as it provides information about the energy used in movement.

Fig. 5.1 shows the length of upper leg bone that have been measured or estimated from fossils of four hominid species (*Australopithecus*, *H. erectus*, *H. neanderthalensis* and *H. sapiens*) covering a time period of approximately three and a half million years.

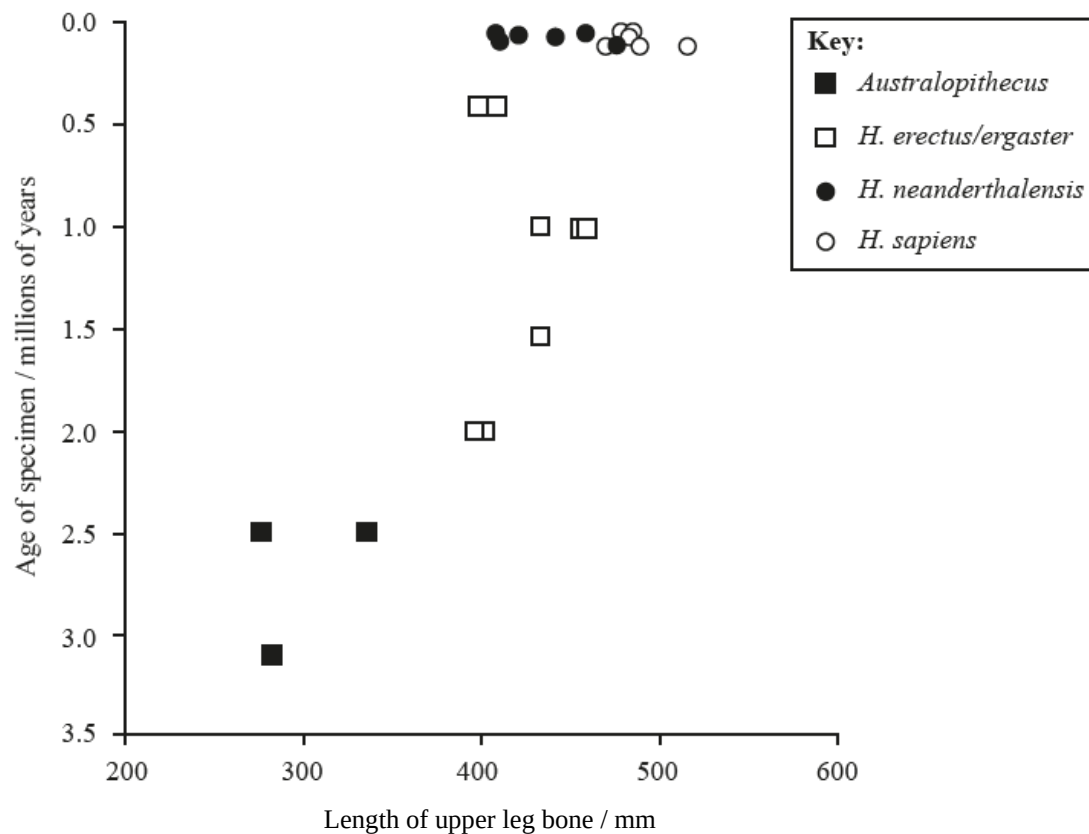


Fig. 5.1

Source: Adapted from K. Steudel-Numbers and M. Tilkens (2004) *Journal of Human Evolution*, 47, pages 95–109.
 ‘The effect of lower limb length on the energetic cost of locomotion: implications for fossil hominins’

- (a) With reference to Fig. 5.1, explain if increasing upper leg bone length provides an evolutionary advantage.

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.....[3]

(b) Suggest one limitation of using fossil records in evolutionary studies.

.....
[1]

We would expect natural selection to eliminate genetic variations with negative effects from human population. However, the study on human genetics reveals that several recessive alleles that are known to cause diseases have been preserved in human populations.

For instance, the sickle-cell allele is carried by half the people in some areas of Africa. This distribution seems to result from the counterbalancing effects of anaemia and malaria, a disease that formally causes high death rates in equatorial Africa.

(c) Account for the high frequency of sickle cell anaemia allele in the African populations.

.....

[3]

(d) Explain why population is the smallest unit that can evolve.

.....

[2]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

1.

(a) Compare aerobic and anaerobic respiration in animals. **[7]**

(b) Both Darwin and Wallace independently came up with the theory of evolution by natural selection. They concluded that organisms had to struggle for survival, and that each subsequent generation slowly became better adapted to the environment.

Using named example, explain how environmental factors act as forces of natural selection. **[8]**

[Total: 15]

2.

(a) Contrast between stem cells and cancer cells. **[6]**

(b) Based on data collected over the years on weather patterns and greenhouse gas emissions, researchers have concluded a causal relationship between greenhouse gas emission and enhanced global warming. Biotic factors like coral reefs are sensitive and vulnerable to changes in climate.

Discuss how weather patterns would be affected and the consequences of such changes, along with how corals reefs are impacted by enhanced global warming. **[9]**

[Total: 15]

End of Paper

[Total: 15]